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Domain of Dependence

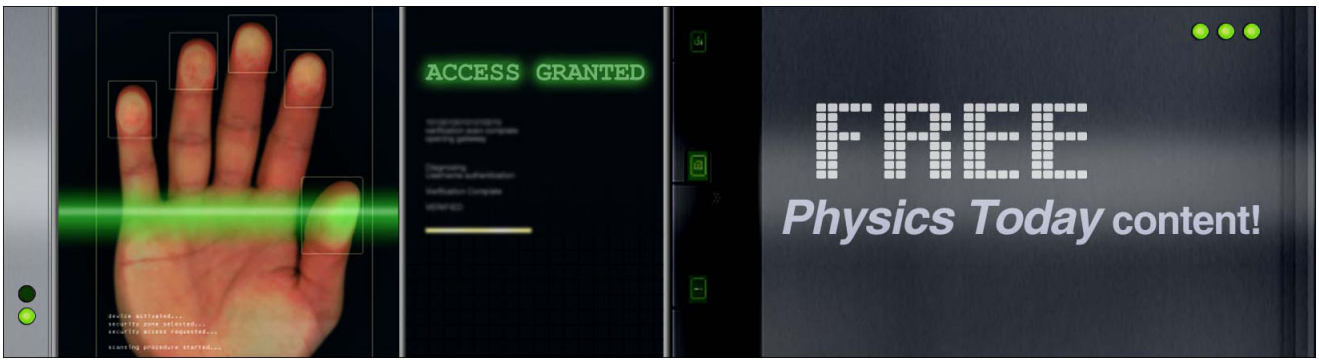
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Domain of Dependence

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The various properties of the domain of dependence (Cauchy development) which have been found particularly useful in the study of gravitational fields are reviewed. The basic techniques for constructing proofs and counterexamples are described. A new tool—the past and future volume functions—for treating certain global properties of space-times is introduced. These functions are used to establish two new theorems: (1) a necessary and sufficient condition that a space-time have a Cauchy surface is that it be globally hyperbolic; and (2) the existence of a Cauchy surface is a stable property of space-times.

1. INTRODUCTION

The concept of the domain of dependence, or Cauchy development, of a spacelike 3-surface has recently found a number of important applications to global problems in general relativity, e.g., in the proofs (and often in the statements) of theorems about singularities[1-6] in the initial-value problem for gravitational and other fields[7] and in various related questions. Roughly speaking, a point p is in the domain of dependence of a surface S if initial data, given on S, completely and uniquely determine the state of any system at p. The domain of dependence and causal structure[4,8,9] must be ranked as the two most important tools presently available for the study of global properties of space-times.

Global hyperbolicity was invented for dealing with hyperbolic differential equations on a manifold.[10,11] A space-time is said to be globally hyperbolic if, given any two of its points, the collection of all timelike or null (causal) curves joining these points is compact (in a suitable topology). This condition requires, intuitively, that there be no "asymptotic regions," "holes," or "singularities" in the region between the points. Global hyperbolicity is the natural condition to impose, for example, to ensure the existence and uniqueness of solutions—in particular, Green's-function solutions—of hyperbolic equations.[10,11] This condition has also been used in the study of gravitational fields, e.g., in the initial-value problem.[12]

The goals of the present paper are, firstly, to summarize those properties of the domain of dependence which have been found useful in the study of gravitational fields and, secondly, to obtain some new results concerning the domain of dependence and its relation to global hyperbolicity.

In Sec. 2 we introduce the three basic definitions: the domain of dependence, the Cauchy horizon, and a Cauchy surface. (The Cauchy horizon consists of certain boundary points of the domain of dependence. A surface is a Cauchy surface if its domain of dependence is the entire space-time.) We illustrate these definitions with seven examples, intended to suggest a number of basic features of the domain of dependence, the Cauchy horizon, and a Cauchy surface and—what appears to be of at least equal importance in this subject—a number of properties that these objects do not possess. In Sec. 3 we prove a selection of results. Using the constructions and counterexamples of Secs. 2 and 3, the reader should have little trouble in deciding, from the almost infinite number of plausible-sounding conjectures, which are true and which are false. Since there are such a large number of useful but comparatively minor properties of the domain of dependence, it is almost impossible to try to remember them all; for this reason the examples and constructions are as important as the statements of theorems. Sections 2 and 3 are a review in the sense that all of the material, if not published before, is known.

In Sec. 4 we define global hyperbolicity and introduce a topology on certain sets of curves.

Global hyperbolicity and the existence of a Cauchy surface—two properties of space-times each widely

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used in its own area of relativity theory—are equivalent: a necessary and sufficient condition that a space–time have a Cauchy surface is that it be globally hyperbolic. The proof, given in Sec. 5, involves the past and future volumes. Because these volume functions appear to be a useful technique for dealing with global properties of space–times, their properties are summarized.

Finally, in Sec. 6 we show that the existence of a Cauchy surface is a stable property of space–times. That is, arbitrary sufficiently small variations in the metric (in a suitable topology) do not destroy the existence of a Cauchy surface.

2. DEFINITIONS AND EXAMPLES

By a *space–time* M we understand a 4-dimensional manifold with a smooth metric g_{ab} of signature $(+, -, -, -)$. We shall also assume that M is time-oriented, that is, that the light cones of M are divided into two systems, past and future.[13]

The fundamental notion is that of the domain of dependence, or Cauchy development, of an achronal set S in M. (A set is said to be *achronal*[9] if no two of its points may be joined by a timelike curve.) It will be convenient to think of S as some spacelike or null 3-surface; in fact, as we shall see later, the domain of dependence is uninteresting except in this case. Roughly speaking, a point p is in the domain of dependence of S if the state of any system at p can be completely specified by initial conditions given on S. Because signals must travel along timelike or null curves, we should expect that the initial data on S would completely determine the situation at p if and only if every such curve from p strikes S. More precisely, *the future domain of dependence of* S, $D^+(S)$, *consists of those points p such that every past-directed timelike curve from p without a past endpoint*[14] *intersects* S.[1,2,6,9,15] The *past domain of dependence*, $D^-(S)$, is defined in a similar way by interchanging the roles of past and future. Finally, the (*total*) *domain of dependence* is the set $D(S) = D^+(S) \cup D^-(S)$.

The following remarks will serve to clarify these definitions. It was necessary, first of all, to include the condition "without a past endpoint" in the definition of the domain of dependence, since otherwise $D^+(S)$ would always coincide with S itself: If q is any point not in S, then one can always find a past-directed timelike curve from q (*with* a past endpoint) which does not intersect S. Secondly, although it might appear to be more appropriate to define the domain of dependence using timelike and null curves rather than just timelike curves, the only effect of such a change, as it turns out, is to eliminate certain boundary points from $D^+(S)$. It will later be seen to be more convenient to have such boundary points included in $D^+(S)$. Finally, the definition of the domain of dependence could be applied equally well when S is not achronal, but, once again, the consequence is to introduce additional complications without adding anything really new.

Note that the domain of dependence involves only the conformal structure of the space–time.[16] That is to say, if we replace the metric g_{ab} by $\Omega^2 g_{ab}$, where Ω is any strictly positive scalar field, then each $D^+(S)$ is unchanged.

Fortunately, it turns out that a handful of counter-examples is sufficient to dispose of many plausible-sounding conjectures concerning the domain of dependence. Thus one of the easiest ways to develop an intuitive feeling for the properties of $D^+(S)$, as well as to check and discover new properties, is through examples. We shall give seven examples here. (In all examples involving Minkowski space, we adopt the usual coordinates t, x, y, and z.)

Example 1: Let M be Minkowski space, and let S be the x axis, i.e., the curve given by $t = y = z = 0$. From any point not on S, we may always find a past-directed timelike curve, without a past endpoint, which fails to intersect S. Thus $D^+(S) = D^-(S) = D(S) = S$. This property would, of course, continue to hold were S any spacelike curve or 2-surface. We shall see in Sec. 3 that, at least locally, S must be a spacelike or null 3-surface if $D^+(S)$ is to contain any points other than those of S.

Example 2: Let M be Minkowski space, and let S be the hyperboloid of revolution given by

$$S = \{(t, x, y, z) \mid t^2 - 1 = x^2 + y^2 + z^2, t < 0\}.$$

It will be seen from Fig. 1 that

$$D^+(S) = \{(t, x, y, z) \mid t^2 \geq x^2 + y^2 + z^2 \geq t^2 - 1, t \leq 0\},$$

$$D^-(S) = \{(t, x, y, z) \mid t^2 - 1 \geq x^2 + y^2 + z^2, t < 0\}.$$

[13] If M were not time-orientable, then we could always find a covering space of M—representing exactly the same physical universe—which is. Many of the results can be applied directly to the non-time-orientable case, but the statements and proofs become somewhat more complicated.

[14] A point q is said to be an *endpoint* of a curve γ if γ enters and remains in every neighborhood of q. Note that q need not necessarily be a point of γ.

[15] R. Courant and D. Hilbert, *Methods of Mathematical Physics* (Interscience Publishers, Inc., New York, 1965), Vol. II.

[16] In fact, the notion of the domain of dependence can be generalized to certain casual spaces. See R. Geroch, E. H. Kronheimer, and R. Penrose, "Ideal Points in Space–Time" (to be published).

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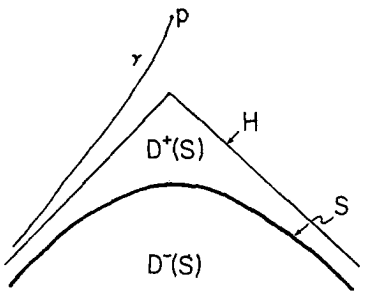


FIG. 1. The domains of dependence of a hyperboloid of revolution in Minkowski space (two spatial dimensions suppressed). Null geodesics in this figure are straight lines inclined at 45° to the vertical. The point p is not in $D^+(S)$ because the past-directed timelike curve γ from p has no past endpoint and does not intersect S.

[From the point $p = (1, 0, 0, 0)$, for example, there is a past-directed timelike curve, γ in the figure, which fails to intersect S, and so $p \notin D^+(S)$.] The only achronal set in M which contains S is S itself, and so we cannot further extend S while keeping it achronal. That is to say, S is a spacelike surface in Minkowski space such that $D(S) \neq M$, and such that S is not included in any achronal set whose domain of dependence *is* M. However, there are other surfaces in Minkowski space—for example, the plane S' given by $t = 0$—such that $D(S') = M$.

Consider, in the above example, the "future boundary" of $D^+(S)$, that is, the past light cone

$$H = \{(t, x, y, z) \mid t^2 = x^2 + y^2 + z^2, t \leq 0\}$$

of the origin. It is easily verified that H is achronal, and that $D^+(H) = H$, $D^-(H) = D(S)$. [Note that if, in our original Minkowski space, the point at the origin had been removed from the manifold, then we should have had $D^-(H) = H$.] More generally, let H be any achronal subset of $D^+(S)$. If p is a point of $D^+(H)$, then every past-directed timelike curve γ, without a past endpoint, from p intersects H. But $H \subset D^+(S)$, and so γ intersects S. Thus $D^+(H) \subset D^+(S)$, and, by a similar argument, $D^-(H) \subset D(S)$.

Example 3: Let M be Minkowski space with the single point at the origin, $O = (0, 0, 0, 0)$, removed, and let S be the closed, spacelike 3-disk

$$S = \{(t, x, y, z) \mid x^2 + y^2 + z^2 \leq 4, t = -1\}$$

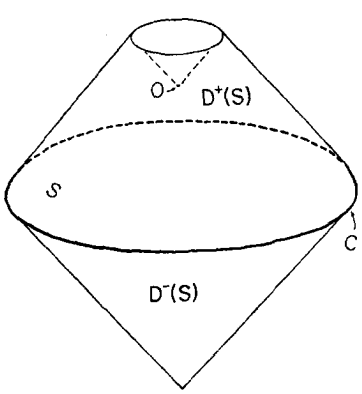


FIG. 2. The domains of dependence of a disk in Minkowski space with the single point O removed (one spatial dimension suppressed). Points to the future of O are not in $D^+(S)$. The set $D^+(S)$ [respectively, $D^-(S)$] includes its future (respectively, past) boundary whether or not the boundary circle C is included in S.

(see Fig. 2). Then, evidently,

$$D^-(S) = \{(t, x, y, z) \mid -3 \leq t \leq -1, x^2 + y^2 + z^2 \leq (t + 3)^2\}.$$

Were it not for the fact that the point O has been removed from Minkowski space to form M, $D^+(S)$ would be just the time-reverse of $D^-(S)$. However, in the example as given, no point p inside the "future light cone" of O can be in $D^+(S)$, for a past-directed timelike curve from p to O has no past endpoint and does not intersect S. (Of course, O cannot be an endpoint of a curve, for O is not a point of the manifold M.) Thus

$$D^+(S) = \{(t, x, y, z) \mid (t - 1)^2 \geq x^2 + y^2 + z^2 \geq t^2, t > 0, \text{ or } (t - 1)^2 \geq x^2 + y^2 + z^2, 0 \geq t \geq -1\}.$$

Suppose now we replace the set S by S', consisting of S with its bounding 2-sphere,

$$C = \{(t, x, y, z) \mid t = -1, x^2 + y^2 + z^2 = 4\},$$

removed. Then

$$D^{\pm}(S') = D^{\pm}(S) - C.$$

That is to say, the domain of dependence of S' differs from that of S only in that the boundary region C is absent. Note that the rest of the boundaries of D^+ and D^- are the same for S as for S'. This example illustrates a characteristic feature of the domain of dependence: D^+ (D^-) always contains its entire future (past) boundary. (These boundaries will later be defined more precisely.)

Example 4: Let M be the (extended) Schwarzschild solution, and let S be the 3-surface given, in the usual coordinates, by $t = 0$ (see Fig. 3). Then $D^+(S)$ consists of all points on S or to the future of S, and $D^-(S)$ consists of all points on S or to the past of S. Thus

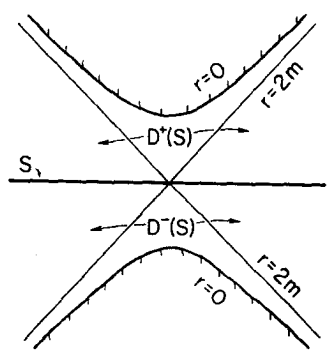


FIG. 3. The domains of dependence of a surface S in the Schwarzschild solution. Each point in the figure (except those on the "singularity," $r = 0$) represents a single "r = const" 2-sphere in the full 4-dimensional space–time. (Thus, for example, the topology of S is $S^2 \times R$.) A curve in the full space–time is represented in the figure by the set of points corresponding to the 2-spheres that curve intersects. Radial null geodesics are represented in the figure by straight lines at an angle of 45° to the vertical, while nonradial null geodesics are curved lines whose angles with the vertical are always less than 45°.

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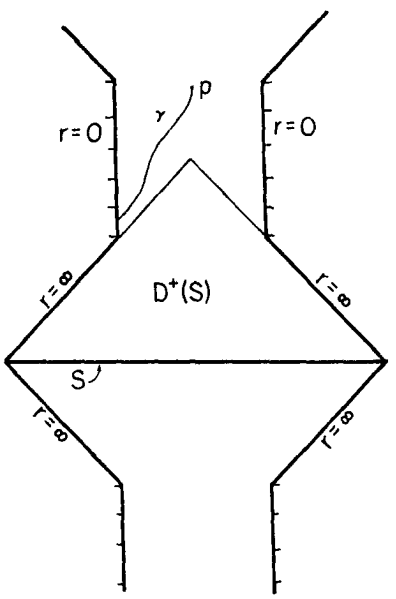


FIG. 4. The future domain of dependence of a surface S in the Reissner–Nördstrom solution. Each point in the figure (except at the "singularity" $r = 0$ and at "infinity," $r = \infty$) represents one of the $r = \text{const}$ 2-spheres in the four-dimensional space–time. A conformal transformation has been applied to bring the asymptotic regions (near $r = \infty$) into the figure. (The same transformation, applied to the Schwarzschild solution, produces a hexagonal-shaped figure.) Radial null geodesics are again straight lines at 45° to the vertical.

$D(S) = M$ and $D^+(S) \cap D^-(S) = S$. [It is always true that $D^+(S) \cap D^-(S) = S$.]

Example 5: Let M be the (extended) Reissner–Nördstrom solution, and let S be the 3-surface given, in the usual coordinates, by $t = 0$ (see Fig. 4). The set $D^+(S)$ is shown in the figure. Points which are sufficiently far up the "throat" between the two asymptotically flat regions, such as p in the figure, are not in $D^+(S)$, for there exists a past-directed timelike curve γ from p which "reaches the singularity at $r = 0$," and so fails to intersect S. In this example, $D(S) \neq M$. In fact, in Examples 3 and 5 there exists *no* achronal set whose domain of dependence is the whole space–time.

Examples 4 and 5 illustrate two ways in which the presence of singularities can influence the domain of dependence.

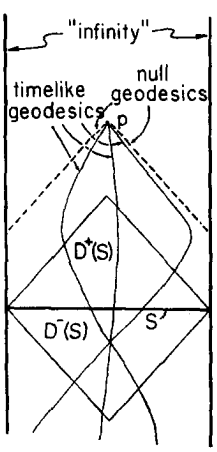


FIG. 5. The domains of dependence of a surface in anti-de Sitter space (two spatial dimensions suppressed). Here "infinity" is represented by a timelike surface, and so certain timelike curves (from p, for example) can reach infinity without first intersecting S. However, every past-directed timelike *geodesic* from p intersects S.

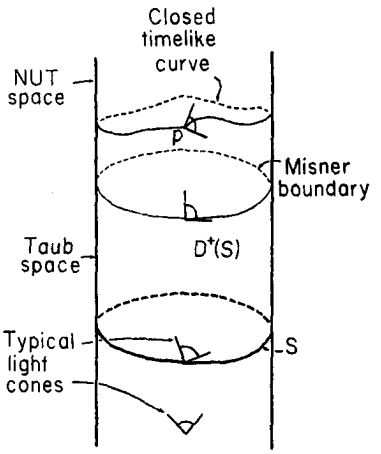


FIG. 6. The future domain of dependence of a surface in Taub–NUT space (two spatial dimensions suppressed: the circles in the figure would be 3-spheres in the full space–time). The metric is invarient under rotations of the cylinder about its axis. Through each point p to the future of the Misner boundary there passes a closed timelike curve, and so no such point may be in $D^+(S)$.

Example 6: Let M be anti-de Sitter space (a space–time, topologically Euclidean, of constant negative curvature), and let S be one of the homogeneous spacelike sections. The sets $D^+(S)$ and $D^-(S)$, as well as a number of typical timelike and null geodesics, are shown in Fig. 5. Note that in this example there exist points, such as p in the figure, such that every past-directed timelike geodesic from p reaches S, but such that $p \notin D^+(S)$. Intuitively, information can come in from "timelike infinity" and affect p without its being registered on S. This example illustrates the point that "timelike curve" cannot be replaced by "timelike geodesic" in the definition of the domain of dependence. (Note also that the null geodesics from p fail to intersect S.)

Example 7: Our final example illustrates the behavior of the domain of dependence in the presence of closed timelike curves. Let M be Taub–NUT space,[17] and let S be one of the compact, homogeneous, nonisotropic spacelike sections in the "Taub part" of the space (Fig. 6). Those points which lie on or below the Misner boundary between Taub and NUT space and on or above S are in $D^+(S)$. Through any point p which lies in the "NUT part" of the space (i.e., above the Misner boundary), there passes a closed timelike curve. The curve which begins at p and which continually describes this closed timelike curve does not strike S and does not have a past endpoint. Thus, no point of NUT space can lie in $D^+(S)$. The above property, evidently, holds in general for any achronal S in any space–time: If p lies on a closed timelike curve, then $p \notin D(S)$. (In the Gödel

[17] E. T. Newman, L. Tamburino, and T. Unti, J. Math. Phys. **4**, 915 (1963); C. W. Misner, J. Math. Phys. **4**, 924 (1963).

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universe, to take a more extreme example, any two points may be joined by a timelike curve. Thus, there does not even exist a nonempty achronal set in the Gödel universe.)

In several of the examples above—e.g., in the Schwarzschild solution and in Minkowski space—we have found an achronal set S such that $D(S) = M$. When this is the case, S is said to be a *Cauchy surface.*[1,2] To say that S is a Cauchy surface for M is a statement about both S and M. Thus, in Example 2, we exhibited an achronal set S in Minkowski space such that $D(S) \neq M$, and such that S is not contained in any Cauchy surface. However, Minkowski space certainly has a Cauchy surface. On the other hand, neither the Reissner–Nördstrom solution, anti-de Sitter space, nor Taub–NUT space have any Cauchy surface.

Intuitively, that S be a Cauchy surface for M means that initial data on S determines the entire evolution of M, past and future.[7] In general, it is not possible to tell, by examining only a neighborhood of S, whether or not S will be a Cauchy surface. Thus, a space-time M (for example, the Reissner–Nördstrom solution) in which it appears, during the early stages of evolution (i.e., near S), that S will be a Cauchy surface may, at some much later time, develop so as to have no Cauchy surface[9]. It is for this reason that the assumption of the existence of a Cauchy surface is a rather unpleasant condition to impose, e.g., in theorems about singularities.

Another useful concept is that of the Cauchy horizon, or "future boundary" of $D^+(S)$. More precisely, the *future Cauchy horizon* $H^+(S)$ of S is defined[4,6] as the collection of all points p of $D^+(S)$ such that no future-directed timelike curve from p intersects $D^+(S)$ other than, of course, at p itself. The *past Cauchy horizon* $H^-(S)$ and *(total) Cauchy horizon* $H(S)$ are defined in analogy with $D^-(S)$ and $D(S)$. We see from Examples 1 and 3 that S may intersect, or even coincide with, its Cauchy horizons.

It follows immediately from this definition that $H^+(S)$ is achronal, and so sets such as $D^+(H^+(S))$, $H^+(H^+(S))$, and $D^-(H^+(S))$ are defined. From the discussion of Example 2, we see that the first two are always identical to $H^+(S)$, while the third must be at least as large as $H^+(S)$ and no larger than $D(S)$.

3. PROPERTIES OF THE DOMAIN OF DEPENDENCE

In this section we shall establish a number of standard results concerning the domain of dependence, Cauchy surfaces, and Cauchy horizons. As most of these results are known and many are fairly easy to prove, we shall not attempt to give the proofs in great detail. It is possible to construct quite a long list of properties of the domain of dependence, and for this reason it is perhaps more feasible to keep in mind a few constructions and counterexamples rather than to try to remember the various properties themselves. Therefore we present here a small selection of results, designed to illustrate certain of the observations of Sec. 2 and to exhibit some of the more important techniques for constructing proofs.

It is convenient to define the *past*[8,9] of a point p, $I^-(p)$, as the collection of points (p excluded) which may be reached from p by a past-directed timelike curve. The *future* of p, $I^+(p)$, is defined in a similar way by interchanging past and future. Note that $I^\pm(p)$ are always open sets. The past (future) of a set is defined as the union of the pasts (futures) of the points of that set. For example, in this notation the definition of the Cauchy horizon takes the form

$$H^+(S) = D^+(S) - I^-[D^+(S)].$$

We saw in Example 1 that, when S is a region of dimensionality less than three in Minkowski space, $D(S) = S$. This property holds in a general space-time:

Property 1: Let p be a point of $D^+(S)$. Then $I^-(p) \cap I^+[S]$ is contained in $D^+(S)$. Furthermore, $I^-(p) \cap S$ is a 3-surface.

Proof: Let $q \in I^-(p) \cap I^+[S]$, and let γ be a past-directed timelike curve from q without a past endpoint. We must show that γ intersects S. Since $q \in I^-(p)$, γ can be extended into the future to p. It follows from the fact that $p \in D^+(S)$ that the resulting curve γ' must intersect S. But $q \in I^+(S)$, and so, since S is achronal, γ' must intersect S at a point to the past of q. Thus, γ intersects S.

To prove the second part of the theorem, choose a nowhere-vanishing vector field t^a on M.[9,18] A trajectory of t^a is defined as a curve, maximally extended, which is everywhere tangent to t^a. There is precisely one trajectory through each point of M, and no trajectory has an endpoint. Since $I^-(p) \cap I^+[S] \subset D^+(S)$, any trajectory which enters the region $I^-(p) \cap I^+[S]$ must continue into the past to intersect S. Furthermore, any point of $I^-(p) \cap S$ must lie on a trajectory which enters the region $I^-(p) \cap I^+[S]$. Let s be a point of $S \cap I^-(p)$, and choose a small neighborhood O of s such that every trajectory which enters O intersects S in O. Since S is achronal, every such trajectory intersects S in the region $O - I^+(s) - I^-(s)$. Therefore,

[18] L. Markus, Ann. Math. **62**, 411 (1955).

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the neighborhood O of s intersects S in an achronal 3-surface.

In the examples of Sec. 2 (especially in Example 3), $D^+(S)$ appears to contain all points which could reasonably be considered as in its "future boundary." This property is necessary to ensure that $H^+(S)$, which is defined as a subset of $D^+(S)$, can be interpreted as this future boundary. Our precise statement asserts that, while $D^+(S)$ itself need not be closed, $D^+(S)$ differs from its closure only in points of the closure of S.

Property 2: $\overline{D^+(S)} = D^+(S) \cup \bar{S}$. In particular, if S is closed, so is $D^+(S)$.

Proof: Since $D^+(S) \supset S$, $\overline{D^+(S)} \supset D^+(S) \cup \bar{S}$. Let p_i be a sequence of points of $D^+(S)$, and let p be an accumulation point of this sequence. We must show that $p \in D^+(S)$ or $p \in \bar{S}$. Suppose that $p \notin \bar{S}$, so that there is some neighborhood O of p which does not intersect S. Let γ be any past-directed timelike curve from p, without a past endpoint. Since the p_i accumulate at p, there is some p_j and some timelike curve γ' from p_j into the past such that γ' joins γ in O and thereafter coincides with γ. But $p_j \in D^+(S)$, and so γ' must intersect S. The intersection cannot occur in O, and so must take place after γ' and γ coincide. Consequently, γ intersects S, and so $p \in D^+(S)$.

Note, however, that it is not true in general that $\overline{D^+(S)} = D^+(\bar{S})$. For example, let S be the surface $t = 0$, $x^2 + y^2 + z^2 > 0$ in Minkowski space.

There are, in fact, a number of relations between the operations of closure, D^+, and H^+. For example, either all or none of the following five sets are closed: S, $D^+(S)$, $D(S)$, $H^+(S)$, and $H(S)$. The proof is along the lines of Property 2, but incorporates the fact that, whenever S is achronal, so is $\bar{S}$, and points of $\bar{S} - S$ can never appear in $D(S)$. One normally considers the domain of dependence only in the case when S is closed. If we think of S as a surface on which to give initial data, then this restriction is not a very severe one, for initial data on S automatically determine, by continuity, the data on $\bar{S}$.

In order that a point p be in $D^+(S)$, we require that every past-directed timelike curve from p, without a past endpoint, intersect S. It might be asked, therefore, what is the effect on $D^+(S)$ if "timelike curve" is replaced by "timelike or null curve." We certainly expect that the domain of dependence, defined in this alternative way, would be no larger than $D^+(S)$. It turns out that neither is $D^+(S)$ made significantly smaller. The only effect of including null curves in the definition of the domain of dependence is to eliminate certain points of the Cauchy horizon from $D^+(S)$.

A continuous curve γ is said to be a *causal curve* if, given any two points p and q of γ and any neighborhood $O \subset M$ of the closed interval $[p, q]$ of γ, there is a smooth curve from p to q in O whose tangent vector is everywhere timelike or null.[19]

Property 3: Let $p \in D^+(S) - H^+(S)$. Then every past-directed causal curve from p, without past endpoint, intersects S.

Proof: Let $\gamma(t)$ be a past-directed causal curve from p, without past endpoint, where t has range $[0, \infty)$ and $\gamma(0) = p$. Let d denote the distance function between pairs of points of M with respect to some positive-definite metric on M. Since $p \in D^+(S) - H^+(S)$, we may choose a point $q \in D^+(S) \cap I^+(p)$ such that $d(p, q) < 1$. Since $\gamma(1) \in I^-(q)$, we may find a past-directed timelike curve $\gamma'(t)$, defined for $t \in [0, 1]$, from q such that

$$\gamma(t) \in I^-(\gamma'(t)), \quad d(\gamma(t), \gamma'(t)) < (1 + t)^{-1} \quad (1)$$

for all $t \in [0, 1]$. Since $\gamma(2) \in I^-(\gamma'(1))$, we may extend the curve γ' so that $\gamma'(t)$ is timelike and subject to (1) for $t \in [0, 2]$. Continuing in this way, we obtain a past-directed timelike curve $\gamma'(t)$, defined and subject to (1) for all $t \in [0, \infty)$. The curve γ' cannot have a past endpoint, for if there were such an endpoint it would [by (1)] also be an endpoint of γ. Therefore, since $q \in D^+(S)$, γ' intersects S. It now follows from (1) that some points of γ are to the past of S, and hence not in $D^+(S)$. Let s be the first point at which γ leaves $D^+(S)$. Since $\overline{D^+(S)} = D^+(S) \cup \bar{S}$, s is either in $\bar{S}$ or in $D^+(S)$. But s cannot lie to the future of S, for in this case $s \in I^-(q) \cap I^+[S] \subset$ interior $D^+(S)$. Therefore, s is in $\bar{S}$. Finally, s must be in S itself, for otherwise a timelike curve, drawn from q and passing through s, would violate the fact that $q \in D^+(S)$.

Example 3 shows that, when $p \in H^+(S)$, the question of whether or not causal curves from p reach S becomes more complicated. For some $p \in H^+(S)$ it may happen that every past-directed causal curve from p intersects S, while this property need not necessarily hold for all points of $H^+(S)$.

We have seen in Sec. 2 that, in many examples in which there is a Cauchy horizon, $H^+(S)$ is a null surface. This property does not hold in general: If, for example, S consists of those points of the plane

[19] In particular, a smooth curve is a causal curve if and only if its tangent vector is everywhere timelike or null. The above definition is rather awkward because it is necessary, for the purposes of Sec. 4, to admit causal curves which are not smooth.

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$t = 0$ in Minkowski space for which x, y, and z are rational, then $S = D^+(S) = H^+(S)$, and $H^+(S)$ is neither a surface nor null. However, $H^+(S)$ always contains null geodesics through its points not included in S. This important property of the Cauchy horizon has been used in theorems about singularities.[4-6]

Property 4[4,7,9]*:* Let $p \in H^+(S) - S$. Then there exists a segment Γ of a past-directed null geodesic from p which remains entirely in $H^+(S)$, and which either has no past endpoint or else has a past endpoint contained in $\bar{S}$.

Proof: Choose a sequence of points p_i which lie to the future of p and approach p as a limit. Since $p \in H^+(S)$, none of the p_i lie in $D^+(S)$. Therefore, through each p_i we may draw a past-directed timelike curve γ_i, without past endpoint, such that the γ_i do not intersect S and, in particular, do not enter $D^+(S)$.

Define a partial limit of the γ_i to be a segment $\Gamma' \subset I^+[S]$ of a past-directed null geodesic from p such that, given any point $q \in \Gamma'$ and any neighborhoods U and V of Γ' and q, respectively, an infinite number of γ_i remain in U, at least until they reach V. We first show that each partial limit is contained in the Cauchy horizon. Let $q \in \Gamma'$. Then each past-directed timelike curve from q enters $I^-(p) \cap I^+[S] \subset D^+(S)$, and so intersects S. Furthermore, no point of $I^+(q)$ is in $D^+(S)$, for otherwise at least one γ_i would enter $D^+(S)$. Thus $q \in H^+(S)$.

Let Γ' be a partial limit, and suppose that Γ' has a past endpoint $q \in I^+[S]$. Choose a small compact neighborhood V of q. The sequence of points at which the γ_i first leave V must, by compactness, have an accumulation point r. Since the γ_i cannot enter $D^+(S)$, $r \notin I^-(q)$. Since the γ_i are timelike and pass arbitrarily close to q, $I^+(r) \supset I^+(q)$. Therefore, there is a segment $\Gamma'' \subset V$ of a null geodesic joining q to r. The tangent vectors to Γ' and Γ'' must agree at q, for otherwise Γ'', and hence the γ_i, would enter $I^-(p) \cap I^+[S] \subset D^+(S)$.[9] Therefore, $\Gamma' \cup \Gamma''$ is a partial limit of the γ_i. We have shown that any partial limit which has a past endpoint in $I^+[S]$ may be extended, as a partial limit, beyond that endpoint.

It follows that there exists a partial limit Γ which either has no past endpoint or else has a past endpoint $s \notin I^+[S]$. In the latter case, since $\Gamma \subset D^+(S)$, we must have $s \in \overline{D^+(S)} = D^+(S) \cup \bar{S}$, and so $s \in \bar{S}$.

Examples 2 and 3 show that there may be more than one null geodesic Γ from p. These same two examples also illustrate the fact that each of the three possibilities admitted by Property 4 (Γ has no past endpoint, Γ has a past endpoint in S, and Γ has a past endpoint in $\bar{S} - S$) can occur.

We next consider some results involving Cauchy surfaces. It is an immediate consequence of the definitions that S is always a Cauchy surface for the space–time given by the interior of $D(S)$, provided that set contains S. Thus all results about Cauchy surfaces apply equally well to the interior of the domain of dependence. We first show that, as expected, the condition that S be a Cauchy surface is precisely the condition that $H(S)$ be empty.

Property 5: Let S be nonempty and M connected. Then a necessary and sufficient condition that S be a Cauchy surface is that $H(S)$ be empty.

Proof: If S is a Cauchy surface, then $D(S) = M$, and so $H(S)$ is empty. Suppose, conversely, that $H(S)$ is empty. In particular, $H(S)$ is closed, and so, therefore, is $D(S)$. We show that $D(S)$ is also open. Let $p \in D(S)$. Since $p \notin H^+(S)$, there is a point r to the future of p in $D(S)$; since $p \notin H^-(S)$, there is a point s to the past of p in $D(S)$. We conclude from Property 1 that the open neighborhood $I^-(r) \cap I^+(s)$ of p is in $D(S)$. Since $D(S)$ is both open and closed, $D(S) = M$.

One often uses Properties 4 and 5 together to establish the existence of a Cauchy surface. By assuming some condition (such as an energy condition) which implies that null geodesics of the type required for Property 4 cannot exist, it follows that $H(S)$ is empty and, therefore, that S is a Cauchy surface.

The following result, using this type of argument, gives a characterization of a Cauchy surface in terms of the behavior of null geodesics.

Property 6[6]*:* A necessary and sufficient condition that an achronal set S be a Cauchy surface is that S be closed and every null geodesic in M intersect and then re-emerge from S.

Proof: If S is a Cauchy surface, then $D(S)$ is closed and so, therefore, is S. By Property 3, every null geodesic intersects S. In the proof of Property 3, it was shown in addition that every causal curve—and, in particular, every null geodesic—re-emerges from S.

Suppose, conversely, that S is closed and that every null geodesic intersects and then re-emerges from S. Since S is achronal, it follows that, for each point r of M, either $r \in S$, or every past-directed null geodesic

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from r intersects S (i.e., $r \in I^+[S]$), or every future-directed null geodesic from r intersects S (i.e., $r \in I^-[S]$). Therefore, since S is closed, any timelike curve which passes from $I^+[S]$ into $I^-[S]$ must intersect S.

The theorem follows from Properties 4 and 5 if we can show that $H^+(S) \cap S$ is empty. Suppose, on the contrary, that there were some point $p \in H^+(S) \cap S$. Every past-directed null geodesic from p emerges from S, and therefore from $H^+(S)$. Consequently, since $H^+(S)$ is achronal, we may, without loss of generality, assume that no point on a past-directed null geodesic from p intersects $H^+(S)$ except, of course, at p itself. [If there were such a geodesic, replace p by the last point in $H^+(S)$ on that geodesic.] Choose a spacelike 2-surface C, topologically a 2-sphere, in a small neighborhood of p such that every past-directed null geodesic from p intersects C, and such that each point of C lies on such a null geodesic. (C is a cross section of the null cone of p.) Each point $q \in C$ is not in $H^+(S)$, and so there exists a q' which is to the future of q and which is in either $I^-[S]$ or $D^+(S)$. Letting C' denote the union of the q', it follows that $I^-[C']$ is open and contains C. Consequently, there exists a point $r \in I^+(p)$ such that every past-directed timelike curve from r enters $I^-[C']$. But $I^-[C'] \subset I^-[S] \cup D^+(S)$, and so every such curve must intersect S. Therefore $r \in D^+(S)$, which contradicts the fact that $p \in H^+(S)$.

The condition "S be closed" is actually necessary in Property 6, as the following example will show. Let S be the surface in Minkowski space given by

$$S = \{(t, x, y, z) \mid t = 1, x^2 + y^2 + z^2 \geq 1, \text{ or } 0 < t < 1, x^2 + y^2 + z^2 = t^2\}$$

(see Fig. 7). Then every null geodesic intersects and re-emerges from S; but, since S is not closed, it cannot be a Cauchy surface. If, in the above example, we remove from the space–time the set of points with $t \leq 0$, then S is closed. (The point at the origin, which was previously in the closure of S, is now no longer a point of the manifold.) However, S is still not a Cauchy surface: Now it does not satisfy the other condition of Property 6—namely, that every null geodesic re-emerge from S.

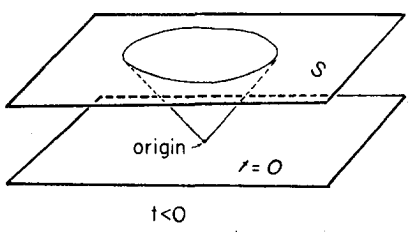


FIG. 7. An achronal set S in Minkowski space (one spatial dimension suppressed). The point marked "origin" is not in S, and so $D^+(S)$ cannot include any point to the future of this point. However, it will be observed that every null geodesic intersects and then re-emerges from S. If, in this space–time, we remove the points on and to the past of the plane $t = 0$, then S is closed. However, certain null geodesics (namely, those which would have passed through the origin) do not now re-emerge from S.

Geodesics are not a very useful way to characterize the domain of dependence. Despite Property 6, it is possible that every past-directed null geodesic from a point p intersects and then re-emerges from S, even though $p \notin D^+(S)$. (Let S be the surface $t = 0$, $x^2 + y^2 + z^2 \geq 1$ in Minkowski space.) In Example 6, every timelike geodesic intersects and re-emerges from S, but S is not a Cauchy surface. If, however, every timelike and every null geodesic intersects S, then S is necessarily a Cauchy surface. In fact, if the point p is such that every past-directed timelike and null geodesic from p intersects S, then $p \in D^+(S)$. The converse is false, e.g., Examples 3 and 6.

Our next property of the domain of dependence asserts that nothing very interesting from a topological viewpoint can take place in $D^+(S)$. Suppose, for example, that we have two universes which, at some time, join together to form a single universe. We might wish to give initial data for this system by specifying a Cauchy surface S consisting of two disjoint pieces, one in each universe. The idea would then be that the initial data on S would predict the separate evolution of the two universes, and also their ultimate confluence. Property 7 asserts, among other things, that such a specification is impossible.

Property 7[6,9]*:* Let S be a Cauchy surface for the space-time M. Then M is topologically $S \times R$. In particular, if M is connected, so is S.

Proof: Choose a nowhere-vanishing timelike vector field t^a on M. Exactly one trajectory of t^a passes through each point of M. Since S is a Cauchy surface, each trajectory intersects S at just one point. We may therefore define a scalar field φ, a "time parameter," by imposing the two conditions $t^a \nabla_a \varphi = 1$ and $\varphi = 0$ on S. It is always possible to rescale t^a so that φ assumes arbitrarily large and arbitrarily small values along each trajectory.[20] Now consider the one-parameter family of surfaces S_φ given by $\varphi = \text{const}$. Exactly one of these 3-surfaces passes through each point of M. Furthermore, any two of these surfaces, S_φ and $S_{\varphi'}$, are homeomorphic: The mapping from S_φ onto $S_{\varphi'}$ is obtained by following along the trajectories of t^a. Hence, $M = S \times R$.

[20] This rescaling may be accomplished as follows: The total length of a trajectory is a lower semicontinuous function on S. Therefore, this function is the limit of some monotonically increasing sequence φ_i ($i = 1, 2, \cdots$) of functions on S. Rescale t^a so that $\varphi = i$ on the surface whose proper distance from S along the trajectories is φ_i.

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Examples 5, 6, and 7 show that the converse of Property 7 is false: M may be a topological product without its having a Cauchy surface. In Sec. 5 we shall obtain a stronger version of Property 7: The one-parameter family of 3-surfaces can be so chosen that each is a Cauchy surface.

Our last two properties of the domain of dependence are proved most easily using a topology on sets of curves to be introduced in Sec. 4. The proofs will be given at the end of that section.

Property 8[2]*:* Let $p \in D^+(S) - H^+(S)$. Then the closure of the set $I^-(p) \cap I^+[S]$ is compact.

Note in addition that when $p \in D^+(S) - H^+(S)$, then $\overline{I^-(p) \cap S}$, a closed subset of $\overline{I^-(p) \cap I^+[S]}$, is also compact. The converse of Property 8 is false. In fact, it is possible for $\overline{I^-(p) \cap I^+[S]}$ to be compact for every $p \in M$ even though S is not a Cauchy surface. (If, however, we assume in addition that M has no closed causal curves, then the converse, as stated above, is true.)

Property 9, which has been used in theorems about singularities, will be required for Sec. 6.

Property 9[2,9]*:* Let $p \in D^+(S) - H^+(S)$. Then there is a timelike geodesic from p to S whose length is at least as large as that of every other smooth causal curve from p to S.

4. GLOBAL HYPERBOLICITY

The notion of global hyperbolicity was introduced by Leray[10] in order to deal with questions of existence and uniqueness of solutions of hyperbolic differential equations on a manifold. It has been used, in particular, in the study of the Cauchy problem in general relativity.[12]

In order to define global hyperbolicity, it is necessary to introduce a topology on certain collections of curves in the space–time M. Let p and q be two points of M, and let $C(p, q)$ denote the collection of all causal curves from p to q. The set of all curves in $C(p, q)$ which lie in O, and O ranges over all open sets in M, defines a basis for a topology on $C(p, q)$. Thus, for example, a sequence γ_i of curves approaches a curve γ if the γ_i are eventually contained in any open set in M containing γ, which corresponds to the intuitive notion of curves "approaching" another curve. It should be pointed out, however, that our topology is not very useful in the presence of closed causal curves. [In particular, $C(p, q)$ is not necessarily Hausdorff in this case.] Fortunately, it is possible to arrange matters so that the topology need only be used in the absence of closed causal curves.[21]

If M has no closed causal curves, then the topological space $C(p, q)$ has a countable basis.[22]

A space–time M is said to be *globally hyperbolic*[10,11] if M has no closed causal curves and if $C(p, q)$ is compact for every p and q. We shall not derive any consequences of global hyperbolicity, since, in the following section, we prove that this condition is equivalent to the existence of a Cauchy surface.

For problems involving Cauchy surfaces, it is convenient to work with a slightly different set of curves. Let p be a point and S a subset of M, and let $C(p, S)$ denote the collection of all causal curves with future endpoint p and past endpoint in S. The set of curves in $C(p, S)$ which lie in O and have past endpoint in O', as O and O' range over all open sets in M, is a basis for a topology on $C(p, S)$. When S is a single point q, the topologies on $C(p, q)$ and $C(p, S)$ coincide.

The interest in $C(p, S)$ stems primarily from the following lemma:

Lemma 10: If S is a Cauchy surface, then $C(p, S)$ is compact.

Proof: Let γ_i be a sequence of curves in $C(p, S)$. Define a partial limit of the γ_i to be a past-directed causal curve Γ from p such that, given neighborhoods U and V of Γ and a point $q \in \Gamma$, respectively, an infinite number of γ_i remain in U at least until they reach V. Suppose a partial limit Γ has a past endpoint q such that, for some neighborhoods U and V of Γ and q, with V compact, only a finite number of the γ_i which remain in U until they reach V have a past endpoint in V. The points at which these γ_i leave V must, by compactness, have an accumulation point $q' \in V$. In this case we may extend Γ to a partial limit with past endpoint q' (see the proof of Property 4). Thus, every partial limit may be extended to a partial limit which either has no past endpoint or else is an accumulation curve of the γ_i. But S is a Cauchy surface, and so any causal curve without past endpoint

[21] Leray's topology is "reasonable" even in space–times with closed causal curves because it refers directly to the mappings which define a curve. Our topology, by referring only to the images of such mappings, is considerably easier to state and visualize, but suffers from the more limited range of applicability described above. In the absence of closed causal curves, the mapping is determined (up to monotonic changes in the parameter) by the image; then Leray's topology and our topology coincide.

[22] This follows from the fact that $C(p, q)$ is a separable metric space. To construct a countable dense set in $C(p, q)$, choose a countable dense set p_i, including the points p and q, in M [see R. Geroch, J. Math. Phys. **9**, 1739 (1968)]. Consider curves $\gamma \in C(p, q)$ which consist of geodesic segments, each of which lies in a normal neighborhood and joins two p_i. A metric on $C(p, q)$ is obtained using the "maximal distance" between curves with respect to a positive-definite metric on M.

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must intersect and re-emerge from S. Such a curve cannot be a partial limit of the γ_i. Thus, every sequence in $C(p, S)$ has an accumulation curve, and so $C(p, S)$ is compact.

Lemma 10 has a number of useful consequences. In particular, it provides us with easy proofs of Properties 8 and 9 of Sec. 3.

To prove Property 8, consider a sequence of points $p_i \in I^-(p) \cap I^+[S]$. For each p_i draw a timelike curve from p to p_i to S. This sequence has a limit curve Γ with future endpoint p and past endpoint in S. Therefore, Γ is a compact set. An infinite number of p_i are contained in any neighborhood of Γ, and so the p_i have an accumulation point. [Note that we have actually proved slightly more than Property 8—namely, that any point of $\overline{I^-(p) \cap I^+[S]}$ lies on a curve in $C(p, S)$.]

To prove Property 9, consider the distance function[23] on causal curves from p to S. This function is upper semicontinuous, and so, since $C(p, S)$ is compact, it assumes its maximal value. The curve on which the distance function is a maximum is the required geodesic. (Similarly, any two points which may be connected by a timelike curve in a globally hyperbolic space-time may also be connected by a timelike geodesic.)

5. GLOBAL HYPERBOLICITY IS EQUIVALENT TO THE EXISTENCE OF A CAUCHY SURFACE

In this section we shall prove our main theorem, the equivalence of global hyperbolicity and the existence of a Cauchy surface. The proof is based on the notion of the past volume and future volume of a point. Select an arbitrary positive volume element dV on M such that the total volume of M is equal to unity[24]:

$$\int_M dV = 1.$$

The *past volume* of a point p, $V^-(p)$, is defined as the volume of the past of p:

$$V^-(p) = \int_{I^-(p)} dV > 0,$$

and similarly for the future. As these volume functions appear to be useful tools for treating certain global aspects of space-times,[25] we first derive some of their elementary properties.

An important property of the volume functions is that, in the absence of closed timelike curves, V^- (respectively V^+) is strictly increasing along each future-directed (respectively, past-directed) timelike curve. [Proof: If there are no closed timelike curves, and if $p \in I^-(q)$, then $I^-(p)$ is a proper subset of $I^-(q)$. Therefore, $V^-(p) < V^-(q)$.] In fact, that V^- be increasing along timelike curves is necessary and sufficient for the absence of closed timelike curves.[26] [Proof: If p and q lie on a closed timelike curve, then $I^-(p) = I^-(q)$, and so $V^-(p) = V^-(q)$.]

In general, V^+ and V^- will not be continuous. For example, let M be Minkowski space with the spacelike half-plane $t = 0$, $x \leq 0$ removed. Then V^- is discontinuous on the null half-plane $0 < t = -x$, and V^+ is discontinuous on the null half-plane $0 > t = x$. (There are similar examples in which only one of the volume functions is discontinuous.)

However, when M is globally hyperbolic, then both V^+ and V^- are continuous everywhere. To prove this result, let p_i be a sequence of points of M, and let p be the limit point of this sequence. We must show that $\lim V^-(p_i) = V^-(p)$. Let ΔV be a subset of M such that $\Delta V \subset I^-(q)$ for some $q \in I^-(p)$. Then, since the p_i approach p, there is some j such that $q \in I^-(p_i)$ whenever $i > j$. That is, the $I^-(p_i)$ eventually contain ΔV. Since $I^-(p)$ can be written as a union of such ΔV's, we conclude that $\underline{\lim}\, V^-(p_i) \geq V(p)$.

We use global hyperbolicity to obtain an upper limit for the $V^-(p_i)$. Let ΔV be a subset of M such that $\Delta V \subset I^+(q)$ for some q such that q cannot be joined to p by a future-directed causal curve. We show that ΔV eventually lies outside of the $I^-(p_i)$. Suppose, on the contrary, that there were an infinite number of p_i such that $\Delta V \subset I^-(p_i)$. Then $q \in I^-(p_i)$ for an infinite number of p_i. Choose any point q' to the future of p. Since the p_i approach p, the p_i are eventually in $I^-(q')$. For each i such that $p_i \in I^-(q') \cap I^+(q)$, draw a timelike curve γ_i from q' to p_i to q. Since M is globally hyperbolic, this sequence of timelike curves from q' to q has a limit curve Γ. Since the p_i approach p, this limit curve passes through p, thus contradicting our assumption that q and p could not be joined by a causal curve. Therefore, ΔV is eventually outside the $I^-(p_i)$. But the interior of $M - I^-(p)$ may be written

[23] A. Avez, Grenoble Universite Institut Fourier **13**, 105 (1963).

[24] Such a volume element always exists. Since M has a metric, it is paracompact [see R. Geroch, J. Math. Phys. **9**, 1739 (1968)] and, therefore, triangulable [see J. H. C. Whitehead, Ann. Math. **41**, 809 (1940)]. Choose a volume element on each of the 4-simplices of M such that the volume of the first simplex is $\frac{1}{2}$, that of the second $\frac{1}{4}$, etc. (No assumption of orientability is necessary.)

[25] S.W. Hawking, Proc. Roy. Soc. (London) **308A**, 433 (1969).

[26] The situation is somewhat more complicated in the case of causal curves. If V^- is strictly increasing along causal curves, then there cannot be closed causal curves. However, it is necessary to impose a condition stronger than the absence of closed causal curves—for example, the strong causality condition (see Ref. 4)—in order to insure that V^- be strictly increasing along causal curves. That is to say, strong causality implies V^- increasing along causal curves which, in turn, implies no closed causal curves. For neither of these implications is the converse true.

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as the union of such ΔV's, and so $\overline{\lim} V^-(p_i) \leq V^-(p)$. We conclude that V^- is continuous on M.

The converse of this result is false: The continuity of both V^- and V^+ is not sufficient for global hyperbolicity (Examples 5 and 6).

Note that the continuity of $V^\pm$ at each point is independent of the choice of volume element. Thus we have an invariant separation of the points of any space–time M into four classes: points at which both V^+ and V^-, V^+ but not V^-, V^- but not V^+, or neither are continuous. We have proven that the collection of space–times for which all points are in the first class includes the globally hyperbolic space–times. It might be interesting to study this collection in more detail.

We now prove our theorem.

Theorem 11: A space–time M is globally hyperbolic if and only if it has a Cauchy surface.

Proof: Suppose first that M has a Cauchy surface S, so that, in particular, M has no closed causal curves. Let p and q be two points of M, and let γ_i be a sequence of past-directed causal curves from p to q.

We consider two cases. If both p and q are in $D^+(S)$, then each curve γ_i may be extended to a curve which intersects S. By Lemma 10, the resulting sequence of causal curves has an accumulation curve Γ. But since each of the γ_i contains q, so must Γ. The portion of Γ between p and q is an accumulation curve of the γ_i. If, on the other hand, $p \in D^+(S)$ and $q \in I^-[S]$, then each of the γ_i must intersect S. Thus, by Lemma 10, the curves $\gamma_i \cap D^+(S)$ have an accumulation curve, say Γ, with past endpoint $s \in S$. Choose a subsequence of the γ_i whose intersection points with S approach s. Then the curves $\gamma_i \cap D^-(S)$, for γ_i in this subsequence, have an accumulation curve Γ'. The curve Γ (respectively, Γ') has a past (respectively, future) endpoint s, and so $\Gamma \cup \Gamma'$ is an accumulation curve of the γ_i. We have shown that, for every p and q, $C(p, q)$ is compact, and therefore that M is globally hyperbolic.

To prove the converse, suppose that M is globally hyperbolic, whence V^+ and V^- are continuous. The function $\lambda = V^-/V^+$ is also continuous and strictly decreasing along each past-directed timelike curve. Let S be the set of points at which $\lambda = 1$. Since λ is strictly decreasing along timelike curves, S is achronal. We shall show that S is also a Cauchy surface. Let p be a point of M with $\lambda(p) > 1$, and let γ be any past-directed timelike curve, without a past endpoint, from p. Since λ is continuous, γ must intersect S, provided we can show that λ assumes arbitrarily small positive values on γ. Let q be a point of M, and ΔV a subset of M in $I^+(q)$. We first show that ΔV cannot be in $I^-(r)$ for every point r of γ. Suppose, on the contrary, that $q \in \bigcap_{r\in\gamma} I^-(r)$. Choose a sequence of points p_i on γ such that $p_{i+1} \in I^-(p_i)$ and such that every point of γ lies to the past of at least one p_i. For each i, draw a timelike curve which begins at p, coincides with γ to p_i, and then continues to q. Since M is globally hyperbolic, this sequence has a limit curve, Γ. The limit curve evidently contains γ. But this is impossible, for γ, if it were contained in a (compact) casual curve from p to q, would then have a past endpoint. Therefore, there must be some point r of γ such that $\Delta V \not\subset I^-(r)$. Since M may be covered by such ΔV's, we conclude that $V^-(r)$ approaches zero as r continues into the past on γ, and, therefore, that γ intersects S. We have shown that every past-directed timelike curve from p intersects S, i.e., that $p \in D^+(S)$. Similarly, if $\lambda(p) < 1$, then $p \in D^-(S)$. Therefore, S is a Cauchy surface.

The proof of Theorem 11 also provides us with a somewhat stronger version of Property 7 (Sec. 3). Consider the one-parameter family of surfaces S_λ in M, given by $\lambda = \text{const}$. Then the S_λ are achronal, and exactly one S_λ passes through each point of M. Furthermore, by the argument used in the theorem, each S_λ is a Cauchy surface. Thus, if a space–time M has a single Cauchy surface, then M may be covered by a one-parameter family of nonintersecting Cauchy surfaces. The argument of Property 7 shows that these Cauchy surfaces all have the same topology.

6. GLOBAL HYPERBOLICITY IS STABLE

In this section we shall prove that global hyperbolicity is a stable property of space–times, i.e., that arbitrary, sufficiently small variations in the metric will not destroy global hyperbolicity. This result is useful both conceptually and as a lemma for simplifying the proofs of various other properties of space–times.[7]

In order to make clear what "sufficiently small" means, we must introduce a topology on the collection $\mathfrak{G}$ of all metrics on M of signature $(+, -, -, -)$. Let $h_{ab}, h'_{ab} \in \mathfrak{G}$. We write $h_{ab} < h'_{ab}$ if every vector which is timelike or null with respect to the metric h_{ab} is strictly timelike with respect to the metric h'_{ab}. Intuitively, $h_{ab} < h'_{ab}$ means that the light cones of h'_{ab} are more "opened out" than those of h_{ab}. The set of $g_{ab} \in \mathfrak{G}$ such that $h_{ab} < g_{ab} < h'_{ab}$, as h_{ab} and h'_{ab} range over $\mathfrak{G}$, forms a basis for a topology on $\mathfrak{G}$. This topology does not distinguish between conformally related metrics: i.e., two metrics which differ only by a conformal factor are always contained in the same open

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sets in $\mathfrak{G}$.[27] This feature is perhaps an advantage in discussing properties of space-times—such as global hyperbolicity—which are conformally invariant. Two metrics which are not conformally equivalent may always be separated by disjoint open neighborhoods.

The characteristic feature of our topology on $\mathfrak{G}$ is that, compared with alternative topologies,[5] it contains a great many open sets. That is to say, it is very difficult in this topology to have a sequence of metrics approach a given metric as a limit. For example, the sequence

$$g_{ab}(n) = \text{diag}\,(\mu_n, -1, -1, -1),$$

where

$$\mu_n = [1 + n^{-2}(x^2 + y^2 + z^2 + t^2)]^{-1},$$
$$n = 1, 2, \cdots,$$

does not approach Minkowski space as a limit. However, the topology is perhaps the most appropriate one for discussing questions of stability.[28]

Using our topology, we define stability as follows: A property of space-times is said to be *stable* if, given any 4-manifold M, the collection of metrics on M which have the given property forms an open set in $\mathfrak{G}$.

Theorem 12: Global hyperbolicity is stable.

Proof: Let M, g_{ab} be a globally hyperbolic space-time. Choose a normalized volume element on M, and let S denote the Cauchy surface given by $V^+ = V^-$. For each value of the parameter τ, set

$$S_\tau = \{p \in M \mid \tau = V^+(p) \leq V^-(p)\}$$

(see Fig. 8). Each S_τ is an achronal set in $D^+(S)$, and exactly one of the S_τ passes through each point of $D^+(S)$.

We first show that, for each value of τ, the closed region $C_\tau = \overline{I^-[S_\tau] \cap I^+[S]}$ between S_τ and S is compact. Let p_i be a sequence of points in C_τ. If, for every $q \in M$, there were only a finite number of $p_i \in I^-(q)$, then we should have $\lim V^+(p_i) = 0$, which contradicts the fact that $V^+(p_i) \geq \tau$ for all $p_i \in C_\tau$. Therefore, there exists a q such that $I^-(q)$ contains an infinite number of p_i. But $\overline{I^-(q) \cap I^+[S]}$ is compact, and so, since it contains an infinite number of p_i, the p_i have an accumulation point. Consequently, C_τ and also S_τ, a closed subset of C_τ, are compact sets.

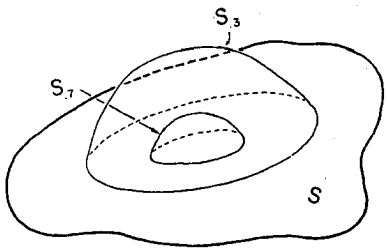


FIG. 8. The Cauchy surface S with the surfaces S_τ in $D^+(S)$. We see that the S_τ form the future boundaries of a nested family of (compact) "bubbles" residing on S. (It may happen that some S_τ are disconnected. Then, as τ decreases, new independent bubbles spring up in other regions of S. Later, as τ decreases further, these bubbles coalesce.)

We say that the family of surfaces S, S_τ are strictly spacelike, with respect to a metric h_{ab}, at a point p if there exists an open neighborhood O of p and a metric $h'_{ab} > h_{ab}$, defined in O, such that no two points of a single surface S_τ or S may be joined by a curve, timelike with respect to h'_{ab} and contained in O. The construction described at the end of the proof shows that S, S_τ may be replaced by surfaces which are strictly spacelike with respect to g_{ab}. (In fact, examples suggest that these surfaces are already strictly spacelike, but no proof is known.) With S, S_τ strictly spacelike with respect to g_{ab}, then, since M is paracompact,[29] there exists a metric $h_{ab} > g_{ab}$ such that S, S_τ are strictly spacelike with respect to h_{ab}.

Let $p \in S_{\tau_0}$, and let γ be a curve from p, past-directed and timelike with respect to h_{ab}, and without a past endpoint. We shall show that γ must intersect S. Since the S_τ are strictly spacelike with respect to h_{ab}, the parameter τ is strictly increasing into the past along γ. Consequently, since C_{τ_0} is compact and since γ has no past endpoint, γ must re-emerge from C_{τ_0}. The boundary of C_{τ_0} consists of S_{τ_0} and certain points of S. The point of emergence of γ cannot be in S_{τ_0}, for τ is strictly increasing along γ. Therefore, γ intersects S.

Using a similar construction for $D^-(S)$, we obtain a metric $h_{ab} > g_{ab}$, defined on M, with respect to which S is a Cauchy surface. Therefore, S is a Cauchy surface with respect to any metric $< h_{ab}$, an open set in $\mathfrak{G}$ which includes g_{ab}. We conclude that global hyperbolicity is stable.

Finally, we outline the construction, required for Theorem 12, which replaces achronal surfaces by strictly spacelike surfaces. Consider two Cauchy surfaces S and S' in a space-time M, and suppose that $S' \subset D^+(S)$. From Property 9 we see that, for each point p, there is a timelike geodesic from p to S whose length is at least as large as that of every other geodesic from p to S. Let $d(p, S)$ denote this length, whence

[27] It is not difficult to construct a similar, but somewhat more complicated, topology which does distinguish between conformally related metrics. The idea of applying this alternative topology to problems in relativity theory was suggested by S. W. Hawking. For a discussion of such topologies, see Ref. 5.

[28] There is another topology on $\mathfrak{G}$ which, because it contains very few open sets, is more suitable for questions involving limits [see R. Geroch, Commun. Math. Phys. 13, 180 (1969).]. It also differs from the topology above in that the manifold need not be given *a priori*.

[29] R. Geroch, J. Math. Phys. 9, 1739 (1968).

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$d(p, S)$ is continuous in p. Define the Cauchy surface

$$\sum(S, S') = \{p \in M \mid d(p, S) = d(p, S'), \quad p \in D^+(S) \cap D^-(S')\}.$$

Let p be a point of $\sum$ with $d(p, S) = d_0 > 0$, γ a timelike geodesic from p to S of length d_0, and q a point on γ such that $d(q, S) = d_0 - \epsilon$. Then, for ϵ sufficiently small, q lies in a normal neighborhood O of p. The set A consists of points in O whose geodesic distance from q is ϵ is a regular, spacelike 3-surface through p. For each point $a \in A$, $d(a, S) \geq d_0$. It follows that $\sum$ is strictly spacelike at all points not included in S or S'.

It is necessary for Theorem 12 to replace S, S_τ by strictly spacelike surfaces. We replace S by the Cauchy surface $\tilde{S} = \sum(S, S')$, where S and S' are the Cauchy surfaces $V^+ = V^-$ and $V^+ = \frac{1}{2}V^-$, respectively. For the S_τ, we first note that for each τ, $W_\tau = (S_\tau \cup \tilde{S}) - I^-[S_\tau \cup \tilde{S}]$ is a Cauchy surface (see Fig. 8). Replace each S_τ by $\overline{\sum(W_{\tau^2}, W_\tau)} - \tilde{S}$.

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Mathematical Description of a System Consisting of Identical Quantum-Mechanical Particles

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In this paper a rigorous description of a system consisting of identical particles is given for which the particle number is a superselection rule. A state of such a system is described by a sequence of density operators $\mathbf{D} = (D^0, D^1, \cdots, D^n, \cdots)$, where D^n acts in n-particle space and the asymptotic behavior is determined by the requirement

$$\sum_{n=0}^{\infty} k^n |D^n| < \infty \quad \text{for all} \quad k = 1, 2, \cdots,$$

where $|\ |$ denotes the trace norm. The (bounded) observables in turn are described by sequences of bounded self-adjoint operators: $\mathbf{B} = (B^0, B^1, \cdots, B^n, \cdots)$ such that

$$\sup_n k^{-n} \|B^n\| < \infty \quad \text{for some} \quad k = 1, 2, \cdots.$$

The expectation value of the observable $\mathbf{B}$ in the state $\mathbf{D}$ can be expressed as

$$\langle B \rangle_D = (\Gamma(\mathbf{B}) \mid \mathbf{D}) = (\mathbf{B} \mid L(\mathbf{D})),$$

where

$$(\mathbf{B} \mid \mathbf{D}) = \sum_{n=0}^{\infty} \mathrm{Tr}\,(B^n D^n);$$

Γ denotes the expansion operator, whereas L stands for its adjoint, the contraction operator. The analog to the n-representability problem, i.e., the so-called representability problem, is put forward and its solution is connected with the solutions of the n-representability problem for different n. Finally, a possible mathematical foundation of the BCS theory is given.

1. INTRODUCTION

In a recent paper (Kummer,[1] subsequently denoted by NRP) the n-representability problem for density operators was considered. The problem can be formulated in the following manner: Let H^1 be the Hilbert space of one particle, H^n the n-fold tensor product of H^1, and $H^{n\wedge}$ the physically relevant subspace (antisymmetric for fermions, symmetric for bosons) of H^n. Given a p-particle density operator D^p, i.e., a positive linear operator in H^p of trace 1, whose range is contained in $H^{p\wedge}$, under what circumstances does there exist an n-particle density operator D^n, such that D^p is the (n, p) contraction of D^n?

[1] H. Kummer, J. Math. Phys. 8, 2063 (1967).

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